

Summary of Findings Tables (Tables S1–S12)

Table S1—Light treatment compared to placebo for ASWPD in adults.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects Light treatment vs. placebo (95% CI)
aMT6s acrophase	43 (1 study ^A)	⊕⊕⊕⊖ LOW ¹ due to imprecision	The mean urinary aMT6s acrophase in the intervention group was 15.00 min lower (92.84 lower to 62.84 higher)
TST (actigraphy)	46 (1 study ^A)	⊕⊕⊕⊖ MODERATE ² due to imprecision	The mean TST(actigraphy) in the intervention group was 34.62 min lower (68.28 lower to 0.96 lower)
TST (subjective)	47 (1 study ^A)	⊕⊕⊕⊖ MODERATE ³ due to imprecision	The mean TST (subjective) in the intervention group was 17.93 min lower (51.13 lower to 15.27 higher)
ISL (actigraphy)	46 (1 study ^A)	⊕⊕⊕⊕ HIGH	The mean ISL (actigraphy) in the intervention group was 1.85 higher (1.45 lower to 5.15 higher)
ISL (subjective)	47 (1 study ^A)	⊕⊕⊕⊖ MODERATE ³ due to imprecision	The mean ISL (subjective) in the intervention group was 41.02 min higher (11.28 lower to 93.32 higher)
SOT (actigraphy)	46 (1 study ^A)	⊕⊕⊕⊖ MODERATE ³ due to imprecision	The mean SOT (actigraphy) in the intervention group was 24.00 min higher (5.03 lower to 53.03 higher)
SOT (subjective)	45 (1 study ^A)	⊕⊕⊕⊖ MODERATE ³ due to imprecision	The mean SOT (subjective) in the intervention group was 19.00 min higher (12.09 lower to 50.09 higher)
SOOffT (actigraphy)	46 (1 study ^A)	⊕⊕⊕⊖ LOW ¹ due to imprecision	The mean SOOffT (actigraphy) in the intervention group was 5.00 min lower (34.19 lower to 24.19 higher)
SOOffT (subjective)	47 (1 study ^A)	⊕⊕⊕⊖ MODERATE ³ due to imprecision	The mean SOOffT (subjective) in the intervention group was 15.00 min lower (44.71 lower to 14.71 higher)
CBT _{Min}	14 (1 study ^B)	⊕⊕⊕⊖ VERY LOW ² due to imprecision	The mean CBT _{Min} in the intervention group was 141.00 min higher (26.10 higher to 255.90 higher)
TST (PSG)	16 (1 study ^B)	⊕⊕⊕⊖ VERY LOW ² due to imprecision	The mean TST (PSG) in the intervention group was 51.30 min higher (2.69 higher to 99.91 higher)
ISL (PSG)	16 (1 study ^B)	⊕⊕⊕⊖ VERY LOW ³ due to imprecision	The mean ISL (PSG) in the intervention group was 11.70 min lower (25.82 lower to 2.42 higher)
SOT (PSG)	16 (1 study ^B)	⊕⊕⊕⊖ VERY LOW ¹ due to imprecision	The mean SOT (PSG) in the intervention group was 15.00 min lower (64.16 lower to 34.16 higher)
SOOffT (PSG)	16 (1 study ^B)	⊕⊕⊕⊖ VERY LOW ³ due to imprecision	The mean SOOffT (PSG) in the intervention group was 29.00 min lower (82.31 lower to 24.31 higher)

Bibliography: ^APalmer et al., 2003; ^BCampbell et al., 1993. ¹CI of the absolute effect crosses the clinical significance threshold (defined by the TF) on both sides of the “no effect” line. ²CI of the absolute effect crosses the clinical significance threshold defined by the TF. ³CI of the absolute effect crosses the clinical significance threshold (defined by the TF) and the “no effect” line. aMT6, 6-sulfatoxymelatonin (urinary metabolite of melatonin); TST, total sleep time; ISL, initial sleep latency; SOT, sleep onset time; SOOffT, sleep offset time; CBT_{Min}, core body temperature minimum; CI, confidence interval; min, minutes; PSG, polysomnography; TF, task force.

Table S2—Melatonin treatment compared to placebo for DSWPD in adults with and without depression.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects
			Melatonin treatment vs. control (95% CI)
TST (PSG) (subgroup with comorbid depression)	28* (2 studies ^{A,C})	⊕⊕⊕⊕ LOW ^{1,2} due to inconsistency, imprecision	The mean TST (PSG) in the intervention groups was 41.44 min higher (13.19 higher to 69.70 higher)
ISL (PSG) (subgroup with comorbid depression)	28* (2 studies ^{A,C})	⊕⊕⊕⊕ HIGH	The mean ISL (PSG) in the intervention groups was 43.52 min lower (52.60 lower to 34.45 lower)
DLMO (min)	11 (1 study ^B)	⊕⊕⊕⊕ LOW ⁴ due to imprecision	The mean DLMO in the intervention group was 55.00 min lower (196.12 lower to 86.12 higher)
TST (actigraphy)	11 (1 study ^B)	⊕⊕⊕⊕ MODERATE ³ due to imprecision	The mean TST (actigraphy) in the intervention group was 15.80 min higher (13.53 lower to 45.13 higher)
ISL (actigraphy)	11 (1 study ^B)	⊕⊕⊕⊕ MODERATE ³ due to imprecision	The mean ISL (actigraphy) in the intervention group was 2.40 min lower (18.08 lower to 13.28 higher)
SOT (actigraphy)	11 (1 study ^B)	⊕⊕⊕⊕ LOW ⁴ due to imprecision	The mean SOT (actigraphy) in the intervention group was 3.00 min higher (130.58 lower to 136.58 higher)
SOFFT (actigraphy)	11 (1 study ^B)	⊕⊕⊕⊕ LOW ⁴ due to imprecision	The mean SOFFT (actigraphy) in the intervention group was 25.80 min higher (102.28 lower to 153.88 higher)
ISL (PSG) (subgroup with no depression)	12* (1 study ^C)	⊕⊕⊕⊕ HIGH	The mean ISL (PSG) (no depression) in the intervention groups was 37.70 min lower (43.65 lower to 31.75 lower)
TST (PSG) (subgroup with no depression)	12* (1 study ^C)	⊕⊕⊕⊕ HIGH	The mean TST (PSG) (no depression) in the intervention groups was 56.00 min higher (48.51 higher to 63.49 higher)

Bibliography: ^AKayumov et al., 2001; ^BMundey et al., 2005; ^CRahman et al., 2010. ¹In studies with crossover design, where more than one measurements were taken on the same patients, all calculations were made based on the number of measurements. ²High level of heterogeneity. ³CI of the absolute effect crosses the clinical significance threshold defined by the TF. ⁴CI of the absolute effect crosses the clinical significance threshold (defined by the TF) on both sides of the “no effect” line. TST, total sleep time; ISL, initial sleep latency; DLMO, dim light melatonin onset; SOT, sleep onset time; SOFFT, sleep offset time; CI, confidence interval; PSG, polysomnography; TF, task force; min, minutes.

Table S3—Melatonin treatment compared to placebo for DSWPD in children/adolescents with no comorbidities.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects
			Melatonin treatment vs. control (95% CI)
DLMO (0.05 mg/kg melatonin)	29 (1 study)	⊕⊕⊕⊕ LOW ¹ due to imprecision	The mean DLMO (0.05 mg/kg melatonin) in the intervention groups was 28.68 min lower (91.15 lower to 33.78 higher)
DLMO (0.1 mg/kg melatonin)	31 (1 study)	⊕⊕⊕⊕ LOW ¹ due to imprecision	The mean DLMO (0.1 mg/kg melatonin) in the intervention groups was 12.35 min lower (74.68 lower to 49.98 higher)
DLMO (0.15 mg/kg melatonin)	32 (1 study)	⊕⊕⊕⊕ MODERATE ² due to imprecision	The mean DLMO (0.15 mg/kg melatonin) in the intervention groups was 33.63 min lower (85.08 lower to 17.83 higher)
ISL (actigraphy, 0.05 mg/kg melatonin)	32 (1 study)	⊕⊕⊕⊕ HIGH	The mean ISL (0.05 mg/kg melatonin) in the intervention groups was 38.39 min lower (58.53 lower to 18.24 lower)
ISL (actigraphy, 0.1 mg/kg melatonin)	33 (1 study)	⊕⊕⊕⊕ HIGH	The mean ISL (0.1 mg/kg melatonin) in the intervention groups was 44.24 min lower (64.44 lower to 24.04 lower)
ISL (actigraphy, 0.15 mg/kg melatonin)	34 (1 study)	⊕⊕⊕⊕ HIGH	The mean ISL (0.15 mg/kg melatonin) in the intervention groups was 43.80 min lower (63.54 lower to 24.06 lower)
SOT (actigraphy, 0.05 mg/kg melatonin)	32 (1 study)	⊕⊕⊕⊕ MODERATE ² due to imprecision	The mean SOT (0.05 mg/kg melatonin) in the intervention groups was 19.92 min lower (48.48 lower to 8.65 higher)
SOT (actigraphy, 0.1 mg/kg melatonin)	33 (1 study)	⊕⊕⊕⊕ MODERATE ³ due to imprecision	The mean SOT (0.1 mg/kg melatonin) in the intervention groups was 33.70 min lower (56.46 lower to 10.95 lower)
SOT (actigraphy, 0.15 mg/kg melatonin)	34 (1 study)	⊕⊕⊕⊕ HIGH	The mean SOT (0.15 mg/kg melatonin) in the intervention groups was 42.77 min lower (63.78 lower to 21.77 lower)

Bibliography: van Geijlswijk et al., 2010. ¹CI of the absolute effect crosses the clinical significance threshold defined by the TF on both sides of the “no effect” line. ²CI of the absolute effect crosses the clinical significance threshold (defined by the TF) and the “no effect” line. ³CI of the absolute effect crosses the clinical significance threshold defined by the TF. DLMO, dim light melatonin onset; ISL, initial sleep latency; SOT, sleep onset time; CI, confidence interval; mg, milligrams; min, minutes; kg, kilograms; TF, task force.

Table S4—Melatonin treatment compared to placebo for DSWPD in children/adolescents comorbid with psychiatric conditions.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects Melatonin treatment vs. placebo (95% CI)
DLMO	132 (2 studies ^{A,B})	⊕⊕⊕⊕ HIGH	The mean DLMO in the intervention groups was 54.22 min lower (76.78 lower to 31.67 lower)
SOT (actigraphy)	130 (2 studies ^{A,B})	⊕⊕⊕⊕ HIGH	The mean SOT (actigraphy) in the intervention groups was 36.57 min lower (56.18 lower to 16.96 lower)
TST (actigraphy)	105 (1 study ^B)	⊕⊕⊕⊖ MODERATE ¹ due to imprecision	The mean TST (actigraphy) in the intervention group was 18.50 min higher (1.15 lower to 38.15 higher)
ISL (actigraphy)	105 (1 study ^B)	⊕⊕⊕⊖ MODERATE ² due to imprecision	The mean ISL (actigraphy) in the intervention group was 18.70 min lower (30.39 lower to 7.01 lower)
TST (subjective)	31 (1 study ^A)	⊕⊕⊕⊖ MODERATE ¹ due to imprecision	The mean TST (subjective) in the intervention group was 29.00 min higher (3.86 lower to 61.86 higher)
ISL (subjective)	33 (1 study ^A)	⊕⊕⊕⊖ MODERATE ¹ due to imprecision	The mean ISL (subjective) in the intervention group was 15.10 min lower (36.64 lower to 6.44 higher)
SOT (subjective)	33 (1 study ^A)	⊕⊕⊕⊖ MODERATE ¹ due to imprecision	The mean SOT (subjective) in the intervention group was 17.00 min lower (57.74 lower to 23.74 higher)
SOfft (subjective)	32 (1 study ^A)	⊕⊕⊕⊖ LOW ³ due to very serious imprecision	The mean SOfft (subjective) in the intervention group was 1.00 min higher (25.30 lower to 27.30 higher)

Bibliography: ^ASmits et al., 2001; ^Bvan der Heijden et al., 2007. ¹CI of the absolute effect crosses the “no effect” line and the clinical significance threshold defined by the TF. ²CI of the absolute effect crosses the clinical significance threshold defined by the TF. ³CI of the absolute effect crosses the clinical significance threshold (defined by the TF) on both sides of the “no effect” line. TST, total sleep time; ISL, initial sleep latency; DLMO, dim light melatonin onset; SOT, sleep onset time; SOfft, sleep offset time; CI, confidence interval; TF, task force; min, minutes.

Table S5—Vitamin B12 compared to placebo for DSWPD in adults.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects Vit B12 vs. placebo (95% CI)
TST (subjective)	50 (1 study)	⊕⊕⊕⊖ VERY LOW ¹ due to imprecision	The mean TST (subjective) in the intervention groups was 19.80 min higher (8.05 higher to 31.55 higher)
SOT (subjective)	50 (1 study)	⊕⊕⊕⊖ VERY LOW ¹ due to imprecision	The mean SOT (subjective) in the intervention groups was 19.00 min lower (32.44 lower to 5.56 lower)

Bibliography: Okawa et al., 1997. ¹CI of the absolute effect crosses the clinical significance threshold defined by the TF. TST, total sleep time; SOT, sleep onset time; CI, confidence interval; TF, task force; min, minutes.

Table S6—Light/combination treatment compared to controls for DSWPD in adults.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Combination treatment vs. control (95% CI)
TST (subjective)	18 (1 study ^B)	⊕⊕⊕⊕ VERY LOW ^{1,2} due to risk of bias, imprecision	The mean TST (subjective) in the intervention groups was 24.60 min higher (49.96 lower to 99.16 higher)
SOT (subjective)	18 (1 study ^B)	⊕⊕⊕⊕ LOW ^{1,3} due to risk of bias, imprecision	The mean SOT (subjective) in the intervention groups was 62.00 min lower (126.88 lower to 2.88 higher)
SOftT (subjective)	18 (1 study ^B)	⊕⊕⊕⊕ VERY LOW ^{1,2} due to risk of bias, imprecision	The mean SOftT (subjective) in the intervention groups was 37.00 min lower (161.64 lower to 87.64 higher)
aMT6s acrophase	45 (1 study ^A)	⊕⊕⊕⊕ MODERATE ³ due to imprecision	The mean aMT6s acrophase in the intervention groups was 40.00 min lower (102.38 lower to 22.38 higher)
SOT (actigraphy)	54 (1 study ^A)	⊕⊕⊕⊕ LOW ² due to imprecision	The mean SOT (actigraphy) in the intervention groups was 0.00 min higher (48.39 lower to 48.39 higher)
SOftT (actigraphy)	54 (1 study ^A)	⊕⊕⊕⊕ LOW ² due to imprecision	The mean SOftT (actigraphy) in the intervention group was 0.00 min higher (43.13 lower to 43.13 higher)
aMT6s acrophase (group with aMT6s acrophase earlier than 6 AM)	22 (1 study ^A)	⊕⊕⊕⊕ LOW ² due to imprecision	The mean aMT6s (group with aMT6s acrophase earlier than 6 AM) in the intervention group was 1.00 min lower (66.62 lower to 64.62 higher)
aMT6s acrophase (group with aMT6s acrophase later than 6 AM)	23 (1 study ^A)	⊕⊕⊕⊕ MODERATE ⁴ due to imprecision	The mean aMT6s (group with aMT6s acrophase later than 6 AM) in the intervention groups was 83.00 min lower (157.91 lower to 8.09 lower)
SOT (actigraphy) (group with aMT6s acrophase earlier than 6 AM)	22 (1 study ^A)	⊕⊕⊕⊕ LOW ² due to imprecision	The mean SOT (actigraphy) (group with aMT6s acrophase earlier than 6 AM) in the intervention group was 13.00 min higher (42.09 lower to 68.09 higher)
SOT (actigraphy) (group with aMT6s acrophase later than 6 AM)	23 (1 study ^A)	⊕⊕⊕⊕ LOW ² due to imprecision	The mean SOT (actigraphy) (group with aMT6s acrophase later than 6 AM) in the intervention group was 26.00 min lower (107.53 lower to 55.53 higher)
SOftT (actigraphy) (group with aMT6s acrophase earlier than 6 AM)	22 (1 study ^A)	⊕⊕⊕⊕ LOW ² due to imprecision	The mean SOftT (actigraphy) (group with aMT6s acrophase earlier than 6 AM) in the intervention group was 7.00 min higher (44.99 lower to 58.99 higher)
SOftT (actigraphy) (group with aMT6s acrophase later than 6 AM)	23 (1 study ^A)	⊕⊕⊕⊕ LOW ² due to imprecision	The mean SOftT (actigraphy) (group with aMT6s acrophase later than 6 AM) in the intervention group was 23.00 min lower (89.12 lower to 43.12 higher)

Bibliography: ^ACole et al., 2002; ^BLack, Bramwell et al., 2007. ¹This is a short paper with lack of detail in methods and results; a lot of information is missing. ²Confidence interval of the absolute effect crosses the clinical significance threshold (defined by the TF) on both sides of the “no effect line.” ³Confidence interval of the absolute effect crosses the “no effect” line. ⁴Confidence interval of the absolute effect crosses the clinical significance threshold defined by the TF. TST, total sleep time; SOT, sleep onset time; SOftT, sleep offset time; aMT6, 6-sulfatoxymelatonin (urinary metabolite of melatonin); CI, confidence interval; TF, task force; min, minutes.

Table S7—Light/combination treatment (multicomponent behavioral interventions + light) compared to controls for DSWPD in children/adolescents.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects Light/Combination treatment vs. control (95% CI)
(Weekdays)			
TST (subjective)	40 (1 study)	⊕⊕⊕⊕ MODERATE ¹ due to risk of bias	The mean TST (subjective) in the intervention groups was 72.00 min higher (37.35 higher to 106.65 higher)
ISL (subjective)	40 (1 study)	⊕⊕⊕⊕ MODERATE ¹ due to risk of bias	The mean ISL (subjective) in the intervention groups was 43.10 min lower (63.74 lower to 22.46 lower)
SOT (subjective)	40 (1 study)	⊕⊕⊕⊕ LOW ^{1,2} due to risk of bias, imprecision	The mean SOT (subjective) in the intervention groups was 42.00 min lower (81.26 lower to 2.74 lower)
SOftT (subjective)	40 (1 study)	⊕⊕⊕⊕ LOW ^{1,2} due to risk of bias, imprecision	The mean SOftT (subjective) in the intervention groups was 23.00 min lower (45.13 lower to 0.87 lower)

Bibliography: Gradisar et al., 2011. ¹Study was not blinded, and an intention to treat analysis was NOT utilized. ²CI of the absolute effect does not reach clinical significance threshold defined by the TF. TST, total sleep time; ISL, initial sleep latency; SOT, sleep onset time; SOftT, sleep offset time; CI, confidence interval; TF, task force; min, minutes.

Table S8—Melatonin treatment compared to placebo for N24SWD in blind adults.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Relative effect (95% CI)
Entrainment (Yes/No)	36 (3 studies)	⊕⊕⊕⊖ LOW	OR 21.18 (3.22 to 139.17)

Bibliography: Lockley et al., 2000; Sack et al., 2000; Hack et al., 2003. CI, confidence interval; OR, odds ratio.

Table S9—Light treatment compared to placebo for ISWRD in elderly with dementia.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects Light treatment vs. control (95% CI)
TST (actigraphy)	35 (1 study ^A)	⊕⊕⊕⊖ VERY LOW ^{1,2} due to risk of bias, imprecision	The mean TST (actigraphy) in the intervention group was 9.00 min higher (67.14 lower to 85.14 higher)
TST (subjective)	24 (1 study ^B)	⊕⊕⊕⊖ VERY LOW ^{3,4} due to risk of bias, imprecision	The mean TST (subjective) in the intervention group was 30.00 min higher (35.63 lower to 95.63 higher)

Bibliography: ^ADowling et al., 2008; ^BMishima et al., 1994. ¹A. No intention-to-treat principle: only completed cases were analyzed. B. Selective reporting: dropouts are not described and some actigraphy variables are not reported (sleep onset, offset, sleep efficiency etc.). ²CI of the absolute value crosses the clinical significance threshold (defined by the TF) on both sides of the “no effect” line. ³A. No blinding, B. No intention-to-treat principle observed. C. Role of control group is vaguely described. ⁴CI of the absolute effect crosses the clinical significance threshold (defined by the TF) on both side of the “no effect” line. TST, total sleep time; CI, confidence interval; TF, task force; min, minutes.

Table S10—Melatonin treatment compared to placebo for ISWRD in elderly with dementia.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects Melatonin treatment vs. placebo (95% CI)
TST (actigraphy) ²⁵	(1 study)	⊕⊕⊕⊖ LOW ¹ due to imprecision	The mean TST (actigraphy) in elderly in the intervention group was 26.23 min higher (84.43 lower to 136.89 higher)

Bibliography: Serfaty et al., 2002. ¹CI of the absolute effect crosses the clinical significance threshold (defined by the TF) on both sides of “no effect” line. TST, total sleep time; CI, confidence interval; TF, task force; min, minute.

Table S11—Melatonin treatment compared to placebo for ISWRD in children/adolescents with neurologic disorders.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects Melatonin treatment vs. placebo (95% CI)
TST (subjective) (min)	34 (1 study)	⊕⊕⊕⊖ MODERATE ¹ due to imprecision	The mean TST (subjective) in the intervention group was 48.45 min higher (6.29 higher to 90.61 higher)
ISL (subjective) (min)	34 (1 study)	⊕⊕⊕⊖ MODERATE ¹ due to imprecision	The mean ISL (subjective) in the intervention group was 51.71 min lower (89.93 lower to 13.49 lower)

Bibliography: Wright et al., 2011. ¹CI of the absolute effect crosses the clinical significance threshold defined by the TF. TST, total sleep time; ISL, initial sleep latency; CI, confidence interval; TF, task force; min, minutes.

Table S12—Comparison of combination of light therapy and melatonin/placebo for ISWRD in elderly with dementia.

Outcomes	# of Participants (studies)	Quality of the evidence (GRADE)	Absolute effects
			Combination treatment vs. control (95% CI)
TST (actigraphy)	32 (1 study)	⊕⊕⊕⊕ VERY LOW ^{1,2} due to risk of bias, imprecision	The mean TST (actigraphy) in the intervention group was 23.00 min lower (101.31 lower to 55.31 higher)

Bibliography: Dowling et al., 2008. ¹A. No intention-to-treat principle: only completed cases were analyzed. B. Selective reporting: dropouts are not described and some actigraphy variables are not reported (sleep onset/offset, etc.). ²CI of the absolute effect crosses the clinical significance threshold (defined by the TF) on both sides of the “no effect” line. PSWS, prescribed sleep-wake scheduling; TST, total sleep time; CI, confidence interval; TF, task force; min, minutes.